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7-Zip with support for Brotli, Fast-LZMA2, Lizard, LZ4, LZ5 and Zstandard

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Code

sebres

CI: improve freeMem9G constraint (win only) - switch to "cat /proc/me...

0bd47d9 · 17 hours ago

github	CI: few use more tests now - hide PA...	4 days ago
Asm	7-Zip Version 26.01 (renormalized)	4 months ago
C	properly handle data error, code revi...	last week
CPP	Add non-posix ACCESSPERMS definit...	2 days ago
DOC	7-Zip Version 26.03 (renormalized, co...	2 weeks ago
DarkMode	Update darkmodelib to 0.64.0	3 months ago
tests	CI: improve freeMem9G constraint (w...	17 hours ago
.gitattributes	added test suite with few initial regre...	last year
.gitignore	Update build flags	10 months ago
.gitmodules	Update darkmodelib to 0.64.0	3 months ago
COPYING	Make License similar to 7-Zip License	5 years ago
README.md	version bump	2 weeks ago

README

This is the Github Page of [7-Zip](#) ZS with support of additional Codecs. The library used therefore is located here: [Multithreading Library](#)

You can install it in two ways:

1. complete setup with additions within the GUI and a modified Explorer context menu
2. only the codec plugin that goes to your existing [7-Zip](#) installation (no GUI changes and no additional Hashers)

Status

release [v26.02-v1.5.7-R2](#) [PayPal](#) [me](#)

Codec overview

1. [Zstandard](#) v1.5.7 is a real-time compression algorithm, providing high compression ratios. It offers a very wide range of compression / speed trade-off, while being backed by a very fast decoder.
 - Levels: 1..22
2. [Brotli](#) v1.2.0 is a generic-purpose lossless compression algorithm that compresses data using a combination of a modern variant of the LZ77 algorithm, Huffman coding and 2nd order context modeling, with a compression ratio comparable to the best currently available general-purpose compression methods. It is similar in speed with deflate but offers more dense compression.
 - Levels: 0..11
3. [LZ4](#) v1.10.0 is lossless compression algorithm, providing compression speed at 400 MB/s per core (0.16 Bytes/cycle). It features an extremely fast decoder, with speed in multiple GB/s per core (0.71 Bytes/cycle). A high compression derivative, called LZ4_HC, is available, trading customizable CPU time for compression ratio.
 - Levels: 1..12
4. [LZ5](#) v1.5 is a modification of LZ4 which was meant for a better ratio at cost of slower compression and decompression. It's superseded by [Lizard](#) now.
 - Levels: 1..15
5. [Lizard](#) v2.1 is an efficient compressor with fast decompression. It achieves compression ratio that is comparable to zip/zlib and zstd/brotli (at low and medium compression levels) at decompression speed of 1000 MB/s and faster.
 - Levels 10..19 (fastLZ4) are designed to give about 10% better decompression speed than LZ4
 - Levels 20..29 (LIZv1) are designed to give better ratio than LZ4 keeping 75% decompression speed
 - Levels 30..39 (fastLZ4 + Huffman) adds Huffman coding to fastLZ4
 - Levels 40..49 (LIZv1 + Huffman) give the best ratio, comparable to zlib and low levels of zstd/brotli, but with a faster decompression speed
6. [Fast LZMA2](#) v1.0.1 is a LZMA2 compression algorithm, 20% to 100% faster than normal LZMA2 at levels 5 and above, but with a slightly lower compression ratio. It uses a parallel buffered radix matchfinder and some optimizations from Zstandard. The codec uses much less additional memory per thread than standard LZMA2.

- Levels: 1..9

7-Zip ZS CLI variants

7z and 7zz provide largely the same core 7-Zip functionality, but they are built/distributed differently (plugin-capable vs. standalone), which can affect available formats/codecs.

Binary	Description
7z	Full 7-Zip command-line tool which loads it's modules/codecs via 7z.so.
7zz	Official standalone 7-Zip binary used on Linux/macOS packages - no external plugins via 7z.so.
7za	Standalone executable which supports fewer archive formats than 7z . (Minimal + LZ4 and Hashes)
7zr	Minimal "light" standalone executable focused on the 7z format. (FLZMA2, Zstd)

7-Zip Zstandard Edition (full setup, with GUI and Explorer integration)

Installation (via setup)

1. download the setup from here [7-Zip ZS Releases](#)
2. install it, like the default [7-Zip](#) one
3. you may check, if the [7-Zip](#) can deal with [Zstandard](#) or other codecs via this command: `7z.exe i`

The output should look like this:

```
7-Zip 26.03 ZS v1.5.7 R1 : Copyright (c) 1999- Igor Pavlov, 2016- Tino Reichardt, 2022- Sergey G. Brester : 2026-09-05
```



Libs:

```
0 c:\Program Files\7-Zip-Zstandard\7z.dll
1 C:\Program Files\7-Zip-Zstandard\Codecs\Iso7z.64.dll
```

Formats:

```
...
0 CK      xz      xz txz (.tar) FD 7 z X Z 00
0         Z      z taz (.tar) 1F 9D
0 CK      zstd    zst zstd tzst (.tar) tzstd (.tar) 0 x F D 2 F B 5 2 5 . . 0 x F D 2 F
B 5 2 8 00
0 C   F      7z      7z      7 z BC AF ' 1C
0     F      Cab     cab     M S C F 00 00 00 00
...
```

Codecs:

```
0 4ED 303011B BCJ2
0 EDF 3030103 BCJ
0 EDF 3030205 PPC
0 EDF 3030401 IA64
0 EDF 3030501 ARM
0 EDF 3030701 ARMT
```

```

0 ED      0 Copy
0 ED      40109 Deflate64
0 ED      40108 Deflate
0 EDF      3 Delta
0 ED      21 LZMA2
0 ED      30101 LZMA
0 ED      30401 PPMD
0 D        40301 Rar1
0 D        40302 Rar2
0 D        40303 Rar3
0 D        40305 Rar5
0 ED      4F71102 BROTLI
0 ED      4F71104 LZ4
0 ED      4F71106 LIZARD
0 ED      4F71105 LZ5
0 ED      4F71101 ZSTD
0 ED      21 FLZMA2
0 EDF      6F10701 7zAES
0 EDF      6F00181 AES256CBC

```

Hashers:

```

0 32      202 BLAKE2sp
0 32      204 BLAKE3
0 4        1 CRC32
0 8        4 CRC64
0 16      205 MD2
0 16      206 MD4
0 16      207 MD5
0 20      201 SHA1
0 32      A SHA256
0 48      208 SHA384
0 64      209 SHA512
0 32      20A SHA3-256
0 48      20B SHA3-384
0 64      20C SHA3-512
0 4        20D XXH32
0 8        20E XXH64

```

My Antivirus X.Y or Virustotal.com say it's malware/virus

In summaray: *It's not maleware* - but it's not my job to chase after these (snake oil manufacturers)

[https://en.wikipedia.org/wiki/Snake_oil].

In detail: you can verify for yourself that the release downloads are generated on the fly by GitHub Actions on this Microsoft-owned platform. If there really were any viruses, they would be generated by machines running in Microsoft data centers 🤖

For example, for the released version v25.01-v1.5.7-R4 :

- it's the one with tag v25.01-v1.5.7-R4
- click on [Adjust Blake3 MAX_SIMD_DEGREE](#)
- scroll down to the Artifacts section
- all the windows binaries are within this ZIP file: 7-Zip ZS Release binaries.zip
- check if the SHA256 hashsums are the same as the files of the release

I started a list of false positive issues [with issue #451](#). Please don't open a new issue. Instead, contact the antivirus vendor and ask them to remove this [false positive](#).

Usage and features of the full installation

- compression and decompression for [Brotli](#), [Lizard](#), [LZ4](#), [LZ5](#) and [Zstandard](#) within the [7-Zip](#) container format
- compression and decompression of [Brotli](#) (.br), [Lizard](#) (.lizard), [LZ4](#) (.lz4), [LZ5](#) (.lz5) and [Zstandard](#) (.zst) files
- handling of ZIP files with [Zstandard](#) compression
- included [lzip](#) decompression support, patch from: <https://download.savannah.gnu.org/releases/lzip/7zip/>
- explorer context menu: "Add to 7z" will use all parameters of the last "Add to Archive" compression dialog (this includes: method, level, dictionary, blocksize, threads and parameters input box)
- squashfs files with LZ4 or Zstandard compression can be handled
- several history settings aren't stored by default, look [here](#) for some info about that, you can restore original 7-Zip behavior via `tools->options->settings`
- these hashes can be calculated: CRC32, CRC64, MD2, MD4, MD5, SHA1, SHA256, SHA384, SHA512, SHA3-256, SHA3-384, SHA3-512, XXH32, XXH64, BLAKE2sp, BLAKE3 (lowercase or uppercase)

```
7z a archiv.7z -m0=zstd -mx0    Zstandard Fastest Mode, without BCJ preprocessor
7z a archiv.7z -m0=zstd -mx1    Zstandard Fast mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=zstd -mx..   ...
7z a archiv.7z -m0=zstd -mx21   Zstandard 2nd Slowest Mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=zstd -mx22   Zstandard Ultra Mode, with BCJ preprocessor on executables

7z a archiv.7z -m0=lz4 -mx0     LZ4 Fastest Mode, without BCJ preprocessor
7z a archiv.7z -m0=lz4 -mx1     LZ4 Fast mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=lz4 -mx..   ...
7z a archiv.7z -m0=lz4 -mx12    LZ4 Ultra Mode, with BCJ preprocessor on executables

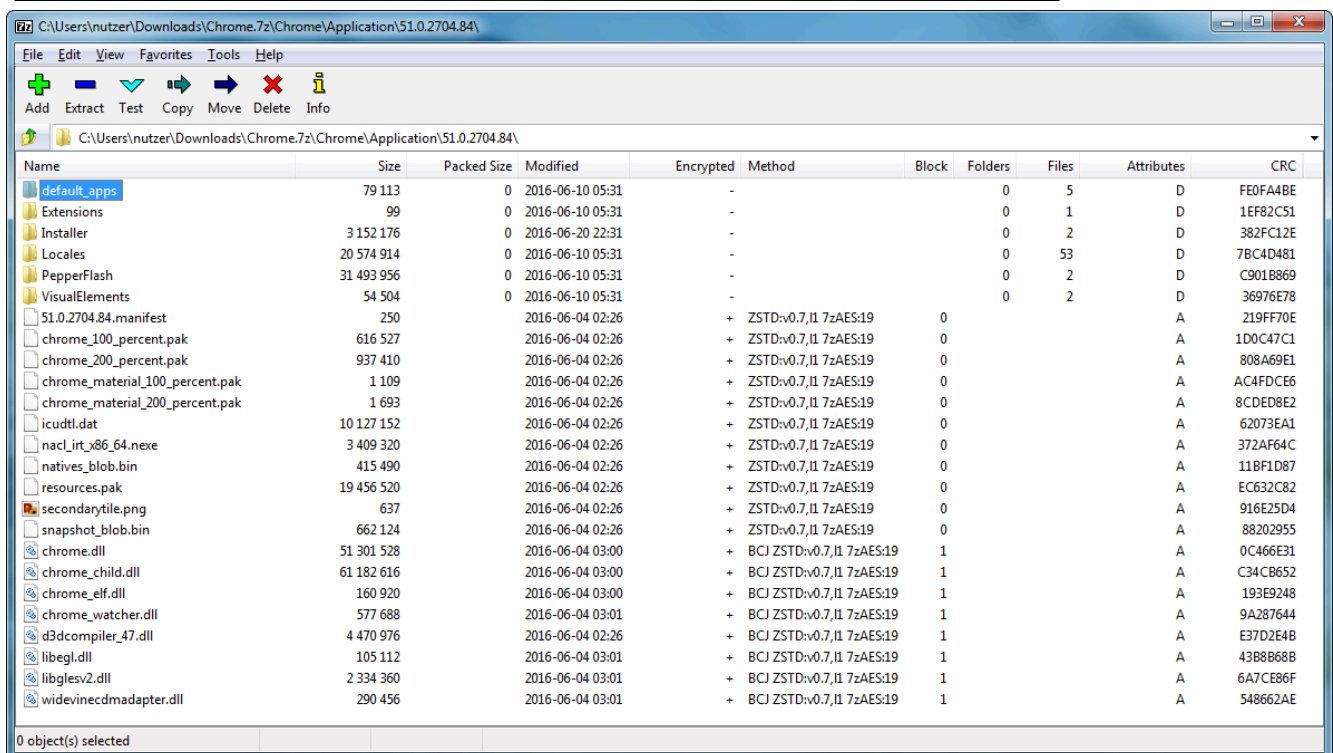
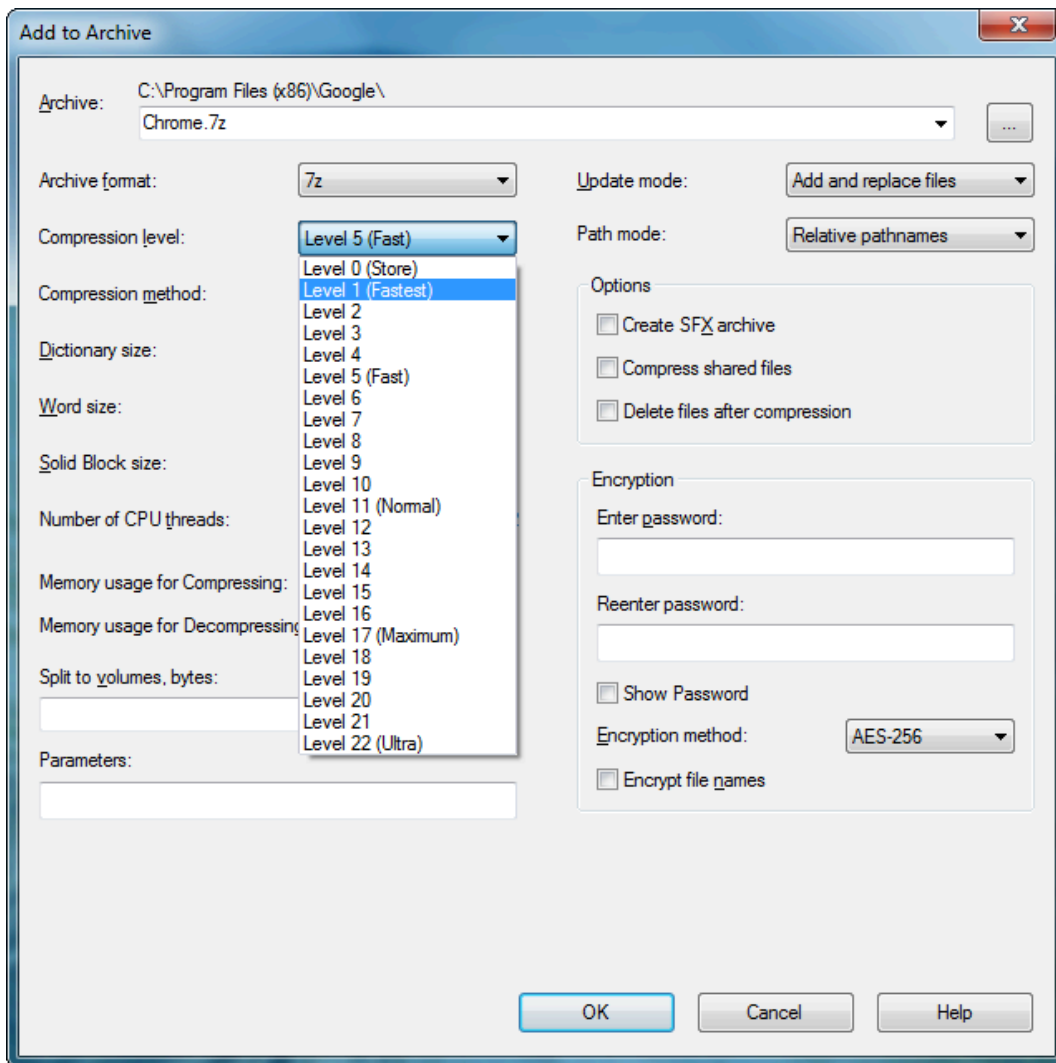
7z a archiv.7z -m0=lz5 -mx0     LZ5 Version 1.5 Fastest Mode, without BCJ preprocessor
7z a archiv.7z -m0=lz5 -mx1     LZ5 Version 1.5 Fast mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=lz5 -mx..   ...
7z a archiv.7z -m0=lz5 -mx16    LZ5 Version 1.5 Ultra Mode, with BCJ preprocessor on executables

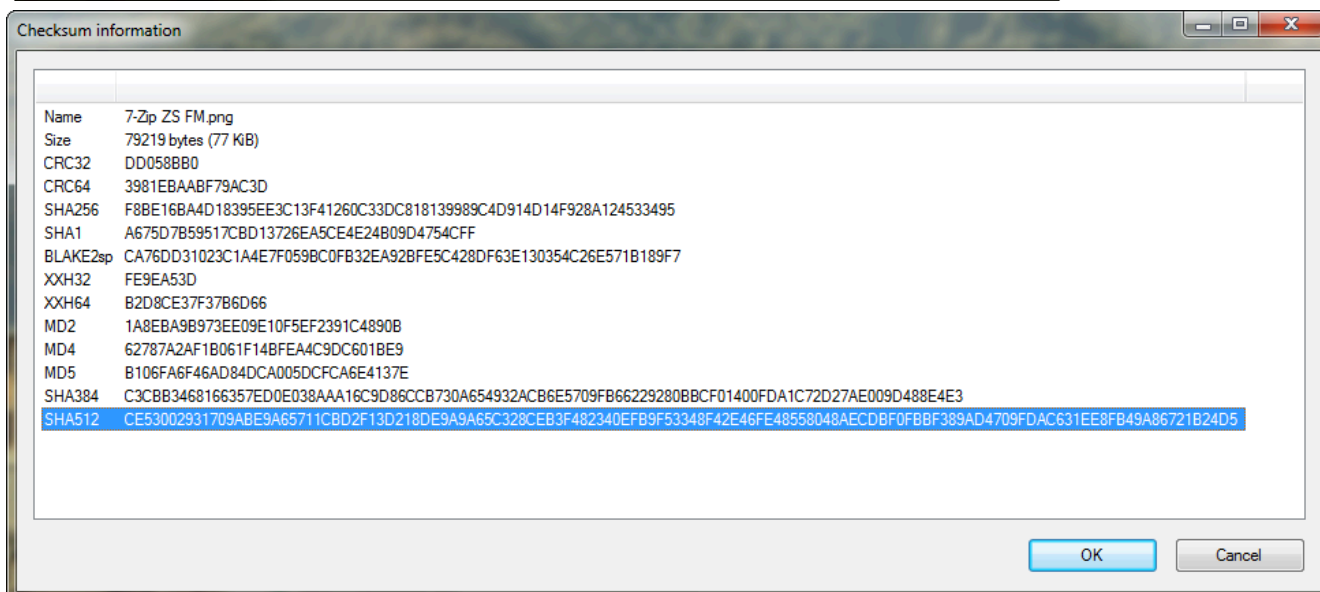
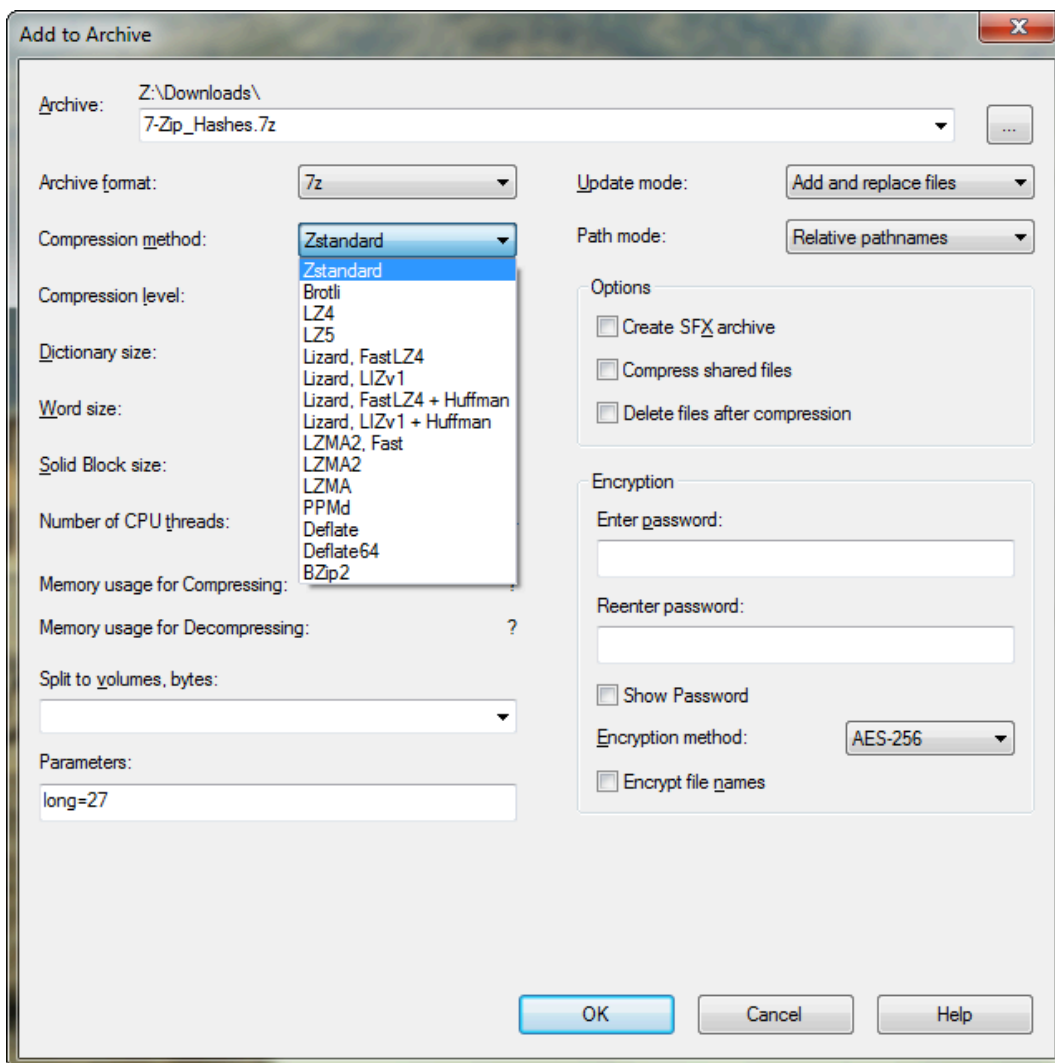
7z a archiv.7z -m0=flzma2 -mx1   Fast LZMA2 Fastest mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=flzma2 -mx.. ...
7z a archiv.7z -m0=flzma2 -mx9   Fast LZMA2 Ultra Mode, with BCJ preprocessor on executables

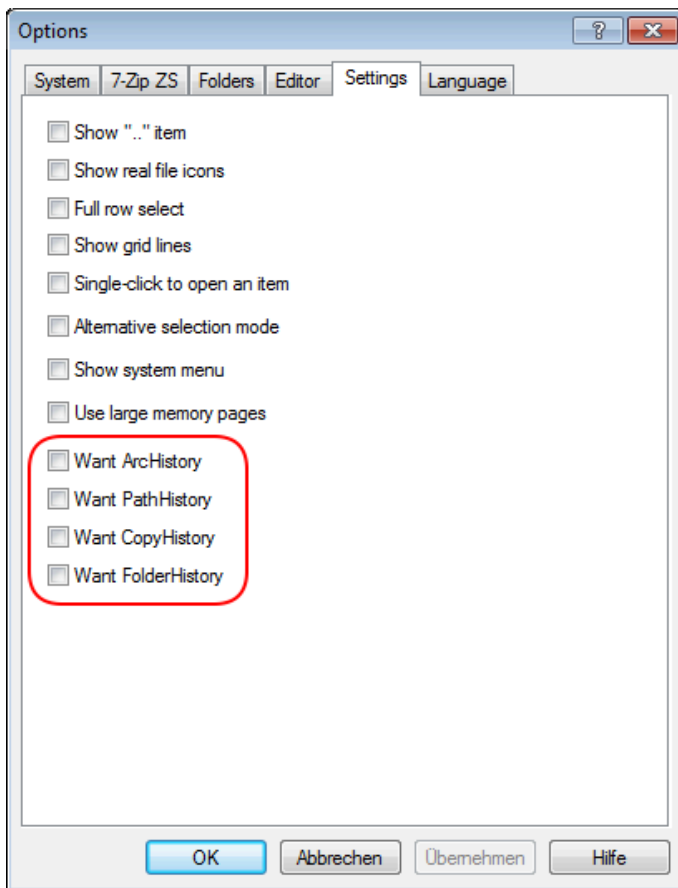
7z x -so test.tar.zst | 7z l -si -ttar
-> show contents of zstd compressed tar archive test.tar.zst

7z x -so test.tar.lz | 7z l -si -ttar
-> show contents of lzip compressed tar archive test.tar.lz
```









Zstandard codec Plugin for Mainline 7-Zip

Installation (via plugin)

1. download the `codecs.7z` archive from here [7-Zip ZS Releases](#), this archive holds binaries, which are compatible with the Mainline version of [7-Zip](#)
2. create a new directory named `codecs` and put in there the `zstd-x32.dll` or the `zstd-x64.dll`, depending on your [7-Zip](#) installation
 - normally, the x32 should go to: "C:\Program Files (x86)\7-Zip\Codecs"
 - the x64 version should go in here: "C:\Program Files\7-Zip\Codecs"
3. you could also replace the `7z.dll` directly within `C:\Program Files (x86)\7-Zip`
4. then you may check if the dll is correctly installed via this command: `7z.exe i`

The output should look like this:

```
7-Zip 21.03 (x64) : Copyright (c) 1999-2021 Igor Pavlov : 2021-05-06
```

```
Libs:
```

```
0 C:\Program Files\7-Zip\7z.dll
```

```
Libs:
```

```
0 c:\Program Files\7-Zip\7z.dll
1 c:\Program Files\7-Zip\Codecs\brotli-x64.dll
2 c:\Program Files\7-Zip\Codecs\flzma2-x64.dll
3 c:\Program Files\7-Zip\Codecs\lizard-x64.dll
4 c:\Program Files\7-Zip\Codecs\lz4-x64.dll
5 c:\Program Files\7-Zip\Codecs\lz5-x64.dll
```



```
6 c:\Program Files\7-Zip\Codecs\zstd-x64.dll
```

```
...
```

Codecs:

```
0 4ED 303011B BCJ2
0 ED 3030103 BCJ
0 ED 3030205 PPC
0 ED 3030401 IA64
0 ED 3030501 ARM
0 ED 3030701 ARMT
0 ED 3030805 SPARC
0 ED 20302 Swap2
0 ED 20304 Swap4
0 ED 40202 BZip2
0 ED 0 Copy
0 ED 40109 Deflate64
0 ED 40108 Deflate
0 ED 3 Delta
0 ED 21 LZMA2
0 ED 30101 LZMA
0 ED 30401 PPMD
0 D 40301 Rar1
0 D 40302 Rar2
0 D 40303 Rar3
0 D 40305 Rar5
0 ED 6F10701 7zAES
0 ED 6F00181 AES256CBC
1 ED 4F71102 BROTLI
2 ED 21 FLZMA2
3 ED 4F71106 LIZARD
4 ED 4F71104 LZ4
5 ED 4F71105 LZ5
6 ED 4F71101 ZSTD
```

Usage (codec plugin)

- compression and decompression for [Brotli](#), [Fast LZMA2](#), [Lizard](#), [LZ4](#), [LZ5](#) and [Zstandard](#) within the 7-Zip container format
- you can only create .7z files, the files like .lz4, .lz5 and .zst are not covered by the plugins
- when compressing binaries (*.exe, *.dll), you have to explicitly disable the bcj2 filter via `-m0=bcj`, when using only the plugin dll's
- so the usage should look like this:

```
7z a archiv.7z -m0=bcj -m1=zstd -mx1 Fast mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=bcj -m1=zstd -mx.. ...
7z a archiv.7z -m0=bcj -m1=zstd -mx21 2nd Slowest Mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=bcj -m1=zstd -mx22 Ultra Mode, with BCJ preprocessor on executables
7z a archiv.7z -m0=bcj -m1=brotli -mxN ...
7z a archiv.7z -m0=bcj -m1=lizard -mxN ...
7z a archiv.7z -m0=bcj -m1=lz4 -mxN ...
7z a archiv.7z -m0=bcj -m1=lz5 -mxN ...
7z a archiv.7z -m0=bcj -m1=flzma2 -mxN ...
```



Codec Plugin for Total Commander

- download [TotalCmd.7z](#)
- install it, by replacing the files `tc7z.dll` and `tc7z64.dll` with the new ones
- you can get detailed information about this [here](#)
- compression and decompression for [Brotli](#), [Lizard](#), [LZ4](#), [LZ5](#) and [Zstandard](#) of the 7-Zip .7z format should work out of the box with Total Commander now :-)

Codec Plugin for Far Manager

- copy the `7z.dll` file from `C:\Program Files\7-Zip-Zstandard\7z.dll` to `C:\Program Files\Far Manager\Plugins\ArcLite\7z.dll`
- then restart the Far manager - and on next start, you will have support for 7-Zip Zstandard archives ;-)

Benchmarks

For benchmarking, I started using the Linux binary `7zz` in 2026.

The test system is an idle Dell PowerEdge R6615 with the following hardware:

- **CPU:** AMD EPYC 9354P (32 cores)
- **Memory:** 128 GB DDR5 (8x 16GB)
- **OS:** AlmaLinux 9 (x86_64)

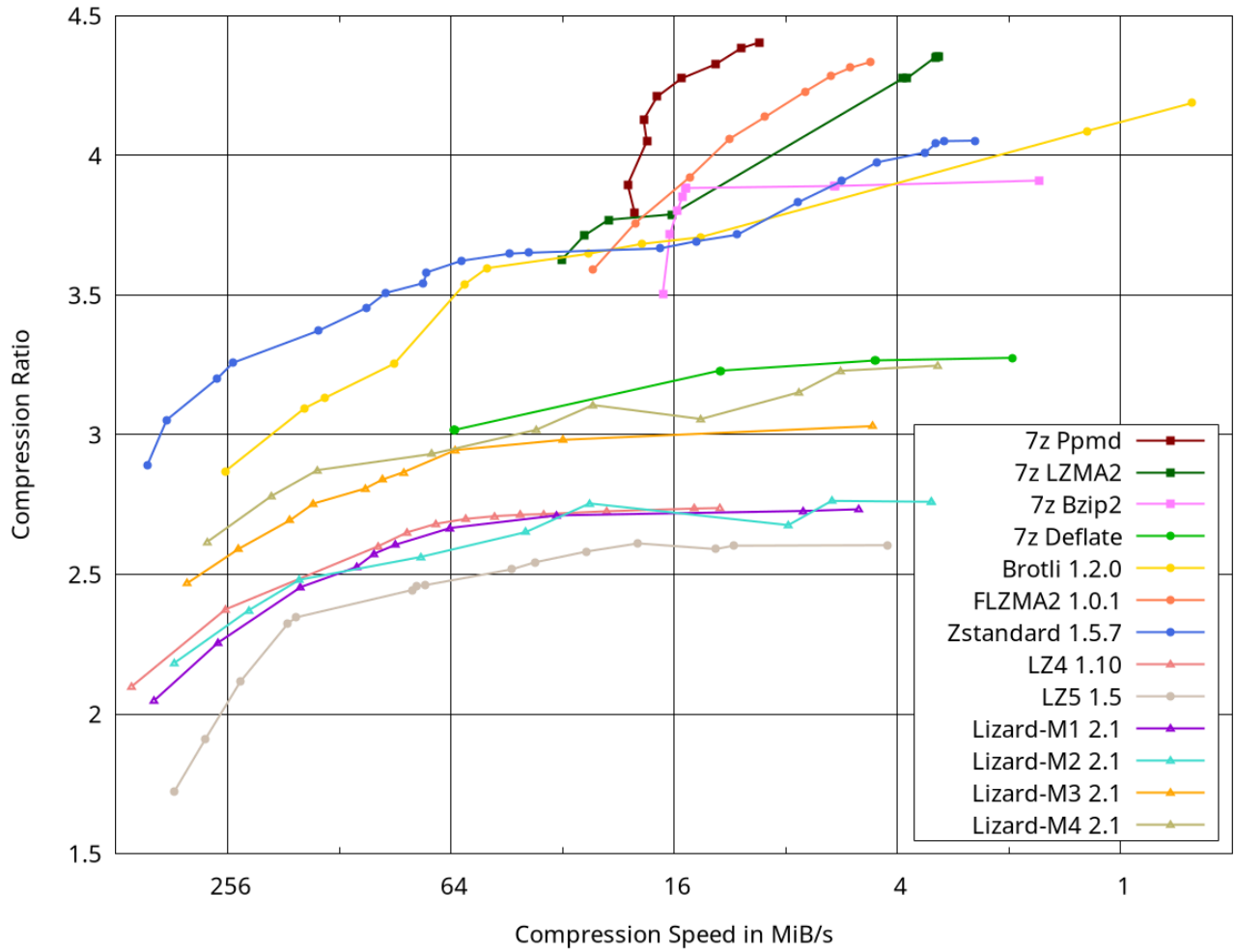
For the tests, the [Silesia compression corpus](#) is used.

Compression tests are performed by running a single-threaded compression for each method: `7z a test.7z -mmt=1 -m0=MethodX`.

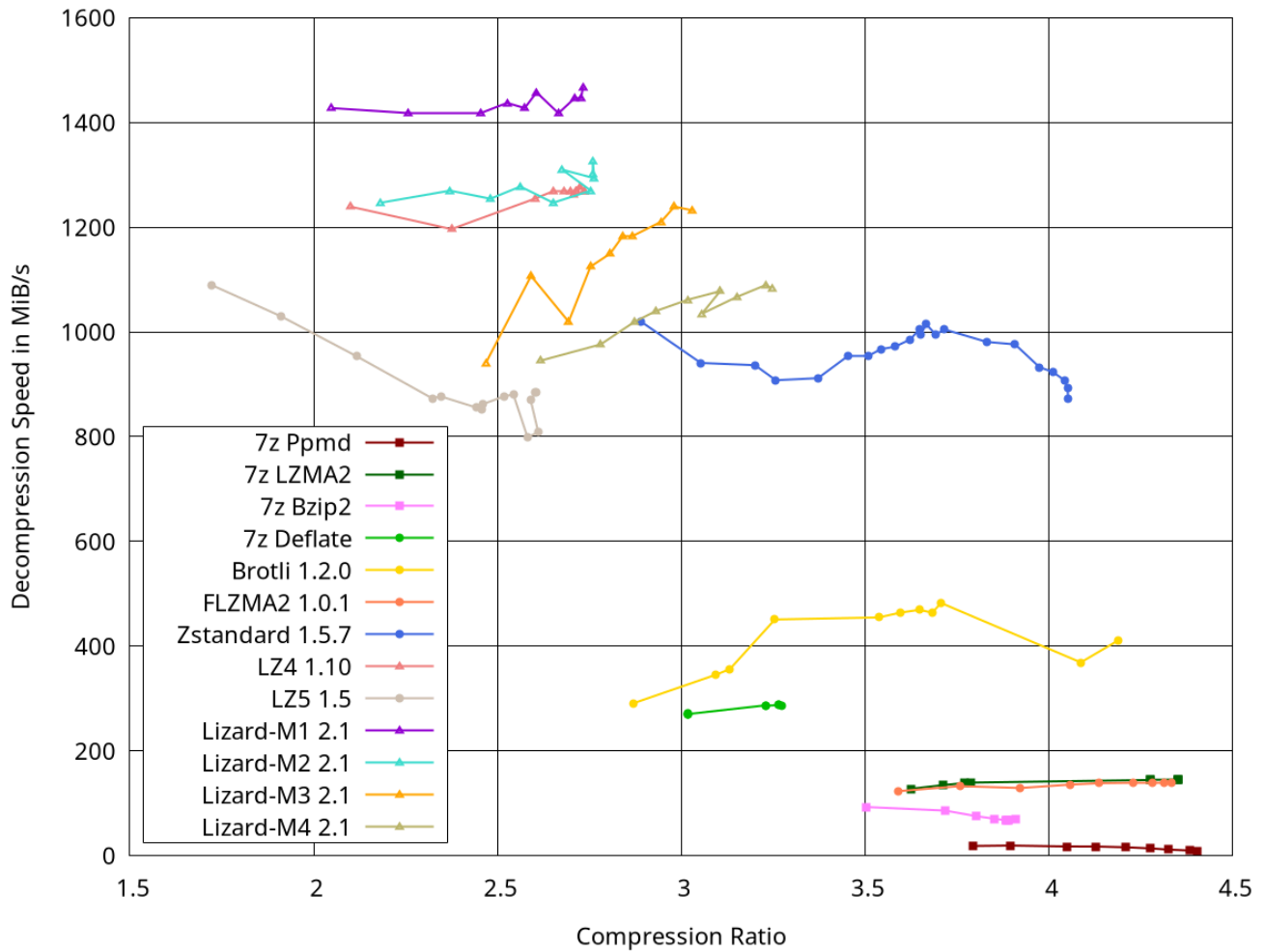
Decompression is tested on the freshly created archive using: `7z t test.7z`. Memory usage and execution times are measured using a [modified GNU time](#). The benchmarks themselves are executed via a [shell script](#).

Results:

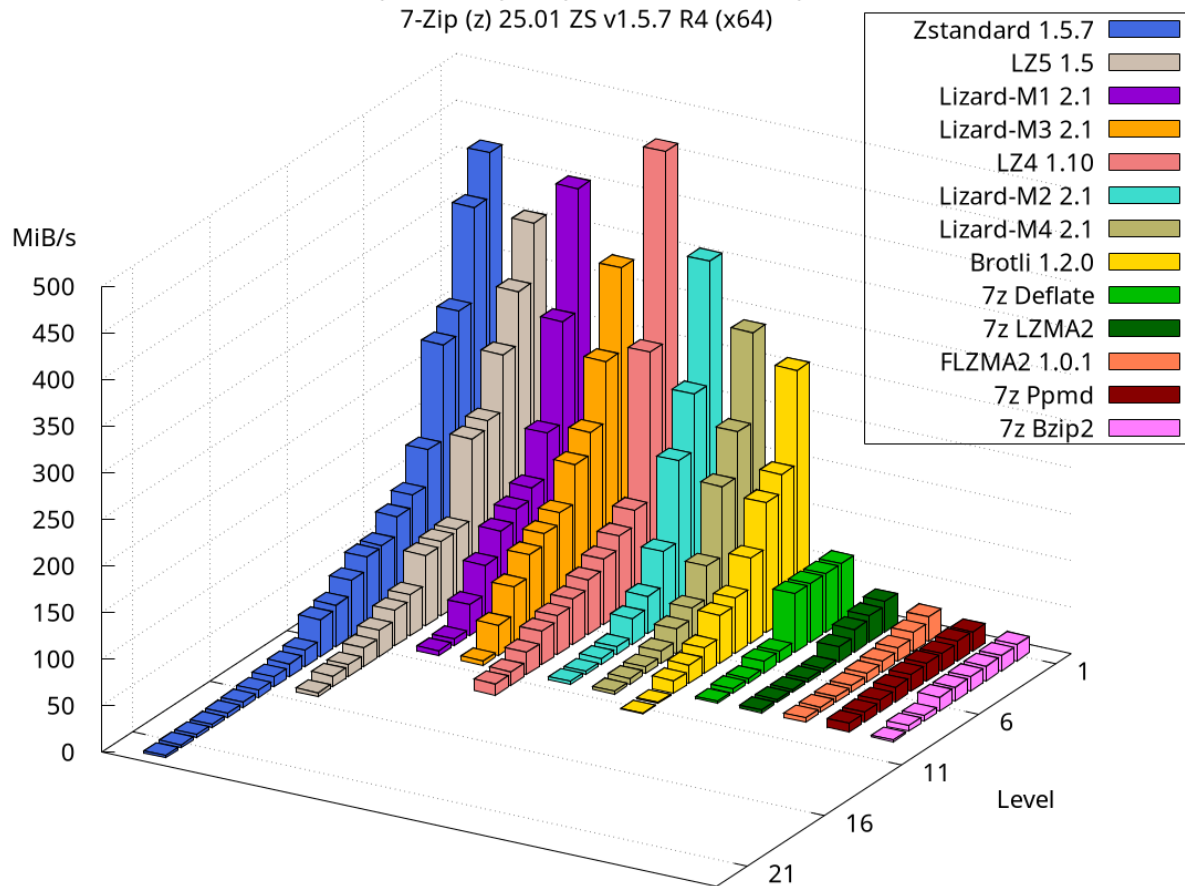
Compression Ratio vs. Compression Speed for 7-Zip ZS Methods [Silesia corpus]
7-Zip (z) 25.01 ZS v1.5.7 R4 (x64)

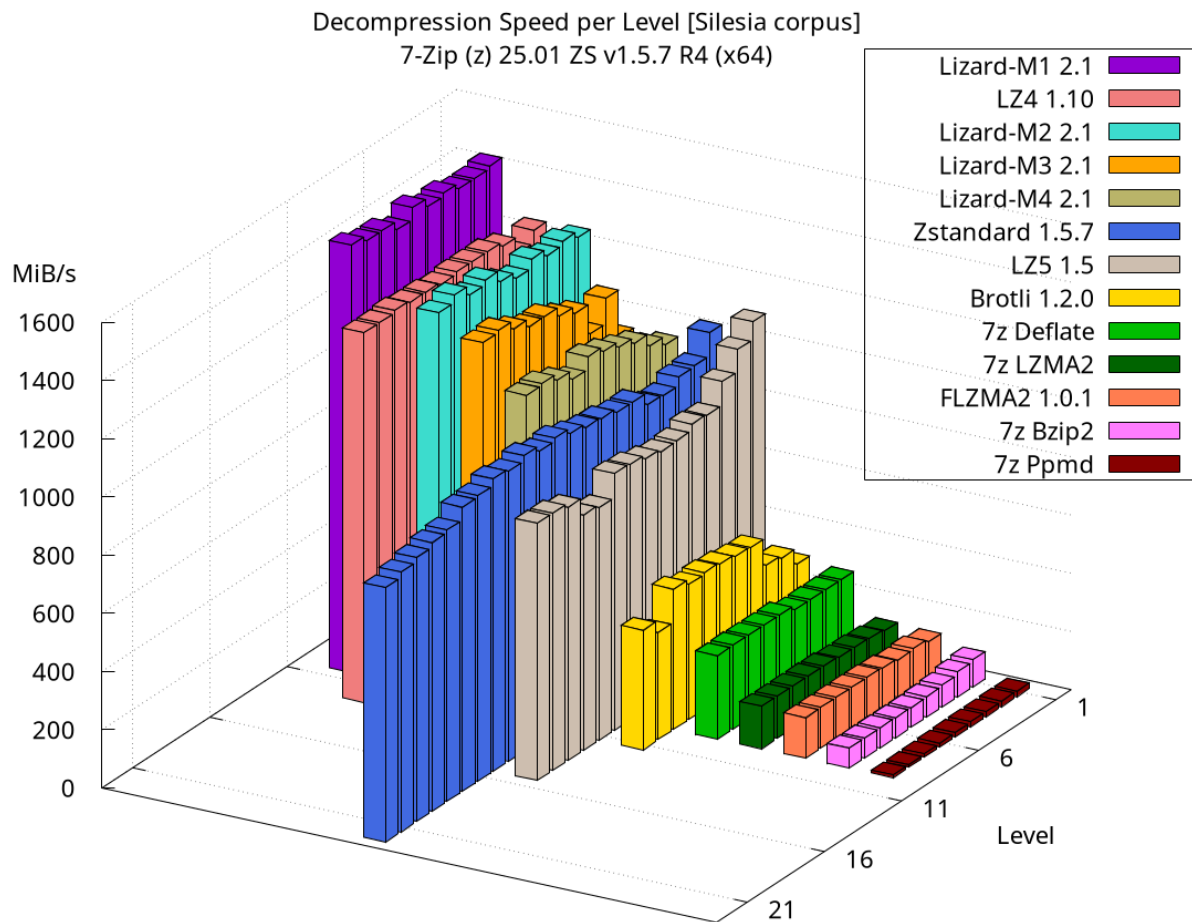


Decompression Speed vs. Compression Ratio for 7-Zip ZS Methods [Silesia corpus]
7-Zip (z) 25.01 ZS v1.5.7 R4 (x64)



7-Zip (z) 25.01 ZS v1.5.7 R4 (x64)





Releases 57

7-Zip ZS 26.02 - v1.5.7 - Release 2 Latest
2 months ago

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